



Field study of thermal comfort in residential buildings in the equatorial hot-humid climate of Malaysia



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ABSTRACT

An extensive field study has been carried out in non-air-conditioned residences in Kota Kinabalu city, in Malaysia for the prediction and evaluation of the effect of the indoor thermal environment on occupants' thermal comfort. A total of 890 responses over one year were gathered. The hot-humid indoor climates of the surveyed buildings have been described and analyzed in terms of air temperature, globe temperature, relative humidity, wind velocity. The clothing insulation ensemble and metabolic rate of the occupants were also characterized. In the aim to assess the comfort temperature in residential buildings in the hot-humid tropics, each analyzed variable was thoroughly compared with the results of two field studies located in the hot-humid tropics, one conducted in Jakarta by Feriadi and Wong [1] and the second in Singapore by de Dear et al. [2]. Multiple and stepwise regressions were applied for the selection of the independent variable for neutral temperature prediction. Air temperature was chosen as an index for the indoor thermal comfort. The comfort temperature was determined using various approaches. The predicted temperature was found to be nearly 30 °C regardless of the adopted approach. The indoor comfort temperature was close to the recorded mean indoor air temperature of all responses having a difference of about 0.7 °C. The mean and the recorded indoor range temperatures seem to have effect in the prediction of comfort temperature.

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1. Introduction

The outdoor climate has always been considered as the main issue in air-conditioned building for the occupants' indoor thermal satisfaction. Prior to the invention of air-conditioning, the ambient temperature was always considered as a normal cycle of change and a normal part of life. While people accept these cycles as normal in nature, they ignore such facts in building thermal design. Passive design principles need to be incorporated and evaluated during the design stage prior to considering any mechanical systems [3]. This is not possible without understanding occupants' thermal comfort toward the indoor thermal environments in their natural daily life.

Several models have been developed during the past years in order to predict human thermal comfort to various climatic conditions [4]. Predicted mean vote (PMV) is among the most well established models widely accepted for building thermal design and field assessment of comfort conditions especially in air-conditioned spaces [5–7]. For naturally ventilated buildings and specifically in warm to hot climates, people however prefer higher

range of indoor air temperatures than that predicted by the PMV model [1,7–9]. Currently, the adaptive approach is widely accepted as efficient tools in predicting indoor comfort temperature for naturally ventilated buildings [6,10,11]. From the adaptive approach perspective, the occupant in a naturally ventilated building is considered a dynamic subject who interacts with the surrounding environment and adjusts to the indoor temperature accordingly; and not a static passive occupant subjected only to external and internal factors. The adaptive approach predicts indoor comfort temperatures of any location from outdoor climatic monthly temperature. One of the adaptive models has been integrated in ASHRAE standard 55 since 2004 [12]. This established model was developed from worldwide thermal comfort field database mainly in naturally ventilated offices [13]. This poses problems when the standard is applied in naturally ventilated residential buildings.

The number of residential buildings included in the development of the adaptive model for summer is less than 8% which contain many locations from the humid to the dry climates. In fact the majority of the databases in the hot-humid tropics were primary from offices. de Dear et al. [13] have noted when comparing between residential and office buildings a distinct differences in the degree of behavioral thermoregulatory adjustment made by residential

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building occupants compared to office workers. For example, seasonal clothing insulation distinctions were more noted in the residential as opposed to office. This may have effect on the comfort indoor temperature. For instance, it has been already shown that people having a higher degree of control over the windows were thermally comfortable at warmer temperatures of 1.5 °C than the group with lower level of control [10]. A difference in neutral temperature of about 1.4 °C also was reported of subjects with desktop task conditioning system compared to those without [14]. The need of worldwide field thermal comfort research in residential buildings also has been reported in many publications [15,16]. Therefore, the research here involves prediction and evaluation of indoor comfort temperature for naturally ventilated residential buildings in a hot-humid tropical environment in the region of Kota Kinabalu city. The specific objectives of this investigation are:

- To investigate and predict the indoor thermal comfort of the occupants in residential buildings in Malaysia.
- To evaluate the validity of the statistical linear regressions models used for indoor comfort temperature prediction.
- To predict comfort temperature based on the mean indoor air temperature of all responses.

2. Methods

The field study took place in Kota Kinabalu city located on the coastline of Sabah state in the year 2007–2008. The latitude of the

location is 5.056' north and the Longitude is 116.03' east. Most of the visited locations are indicated in the black spots in Fig. 1. A summary of the surveyed buildings is listed in Table 1. The thermal comfort field survey was conducted under various indoor climatic conditions and spanned over a period of one full year. The initial sample consisted of 949 records. The obtained data were filtered against the criteria established prior analysis. Only healthy acclimatised adults of 15 years old and above were considered. Records also were reduced to 890 due to inconsistent responses, missing records. The minimum and maximum values of each variable were checked to ensure that all variables are valid. Outliers were discarded with precaution. At least 20–25 min in occupants' residences were used to allow all sensors to achieve final equilibrium with their surroundings, to explain the questionnaire, and to allocate enough time to respondents to fill in all the necessary information and questions that do not require environmental measurements. This also provided sufficient time for subject body precondition [1]. In this paper subjects' response to thermal sensation on the seven-point ASHRAE scale was considered.

In the demographic survey, the age range of occupants was large from 15 to 80 years. There were fewer participants above the age of 51–60 and the highest relative frequency is for the range of 20–30 years. Nearly 42% of the participants were males and 58% were females. The estimated average surface area of the subjects under investigation was 1.62 m². This is more representative of a Malaysian body proportion rather than the worldwide average surface of 1.80 m². The average Dubois body surface of Thailand



Fig. 1. Sites locations of the surveyed buildings.

Table 1
Surveyed buildings.

Type of building	Building material	Number of respondents
Terrace	H.W.	323
Semi-detached and detached houses	Not urban Urban L.W. and/or H.W. H.W.	121 71
Apartment – flat	H.W.	375
Total		890

Note: H.W.: heavy weight building material such concrete or brick; L.W.: light weight building material mostly wooden houses.

population was 1.56 m² [17] and the estimated value for the Indonesian subjects was 1.60 m² [18].

3. Results

3.1. Analysis of personal and indoor environmental parameters

The responses from thermal comfort questionnaire were entered in Excel for a primary analysis then they were reevaluated for careful quality assurance. Detailed descriptive statistics were performed on environmental measurements and personal records. In addition, outcomes from this investigation were compared with other studies carried out in the hot-humid tropics. This offered further insight about similarities and differences of the parameters under investigation in which enable researchers to understand some of the reasons that may led to different outcomes in the determination of comfort temperature.

3.1.1. Clothing insulation

The mean clothing ensembles of the subjects and the standard deviation were 0.27 clo and 0.09 clo, respectively, whereas the range of the clothing ensembles was from about 0.09 clo to 0.56 clo. Overall the results are consistent with the finding of Feriadi and Wong [1]. The authors found that the mean value of occupants' clothing insulation ensembles was about 0.30 clo. A close observation in the de Dear database [19] for the Singapore case study [2] showed slightly higher mean clothing insulation ensembles of 0.38 clo. Both studies added 0.15 clo to take into account the omitted chair insulation from their datasets.

The chair insulation was often omitted in thermal comfort field studies [20]. The value of 0.15 clo is generally added by some researchers when the chair is omitted in their investigations. This value is more representative for a typical executive chair in ISO 7730 and ASHRAE 55 [7,6]. It is valid for clothing ensembles of 0.5 clo to 1.2 clo in ASHRAE 55 [6]. For net chair, metal chair, or wooden side arm chair no additional insulation values were stipulated in the earlier versions of ASHRAE 55 prior 2004. In the present study, the chair insulation was recorded for each subject during the time of the survey and then it was added to the entire clothing ensemble. For the person sitting on the floor, wooden stool, net/metal chair, no additional insulation values were added following ISO 7730 [21]. This was applied as well whenever the chair was judged to have less effect on the clothing insulation ensemble and occupant thermal sensation. In terms of calculation procedure, the current study is likely to be relatively more precise, however, not accurate. In the conducted field study many occupants were bare feet and were not leaned toward the chair when they were filling the questionnaire and some were just sitting on the floor. Additionally, people in their dwellings compared to offices or classrooms have a tendency to wear freely any type of clothes and therefore any standard cannot predict even the type of

clothes worn by occupants in their residences and let along their insulation values. This should be emphasized. Our estimation is no more than a rough approximation because of the limitations stated above.

The results show that the average clothing insulation values from the three studies are less than that assumed in the standards in naturally ventilated buildings for the summer season. This agreement lends support that the value of 0.5 clo adopted generally by standards might not be representative of clothing insulation ensembles of occupants in naturally ventilated residences in the hot-humid tropics. Most probably, the situation is different for the case of offices or classrooms. This observation underscores the errors involved when applying the ASHRAE 55 [6] charts for evaluation purpose as they are valid for clothing insulation ensembles between 0.5 clo to 1 clo only. Again, the insulation values of the chairs as discussed above are valid for clothing insulation of at least 0.5 clo to 1.2 clo as stipulated in the ASHRAE 55 [6] and there is no better substitution elsewhere. From the analyzed results, it is clear that the clothes worn in the hot-humid tropics are very light reflected by the small clothing insulation values.

People in moderate climates wear different clothing ensembles according to the season; lighter in summer and heavier in winter. This study offered the possibility to check the relationship between air temperature and clothing insulation in the hot-humid tropics. Again, the de Dear database [13] offered a great opportunity to compare the results of their study (Fig. 2a) [2] conducted in Singapore with this study (Fig. 2b). The graphs in Fig. 2 indicates that there is no correlation between clothing insulation and indoor air temperature variation as the line-of-best-fit for both studies is approximately an horizontal curve, bearing no trend. This can be observed by the very large scatter of the clo values with the indoor air temperature.

It can be inferred that clothing adjustments to the indoor air temperatures are less likely to occur in Malaysian and Singapore dwellings. This can be explained in part by the narrow and constant range of the outdoor air temperature in the hot-humid tropics all year round. The other most likely explanation is that clothes worn by occupants have already reached the minimum possible level and further decrease in clothing insulating will not took place regardless of the increase in the indoor air temperature. This observation does not imply that under different climate and culture, clothing worn by occupants will not be an important factor in behavioral adjustments to the indoor air temperature. For instance, the coefficient of determination found in the field study carried out in Tunisia was 0.5 for the indoor globe temperature range of 4.5–39.5 °C [37]. The

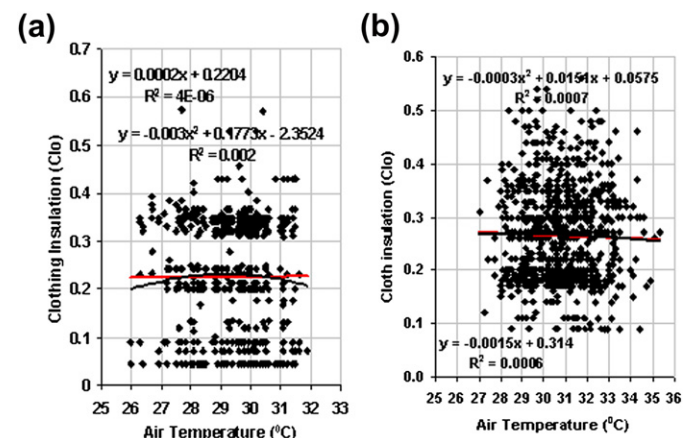


Fig. 2. Correlation of clothing to indoor air temperature.

robust correlation between these two variables compared to the present study can be explained by the very large range of the indoor air temperature in the Tunisia context compared with the narrow range of air temperature in the hot-humid tropics. Humphreys et al. [22] also found from the analysis of nine field surveys that the rate of change of clothing insulation with air temperature occurred in the range 18–28 °C, while below 18 °C and above 28 °C the rate of change was not observed. The means clothing insulation ensembles in their investigation were from 0.4 clo to 1.6 clo. Interestingly, the additional chair insulation of +0.15 clo does not improve the correlation between indoor air temperature and clothing insulation as the slope and the coefficient of determination of the regression remain the same but does affect the intercept and it has effect in the estimation of the PMV. The mean clothing insulation value found in Busch [17] study in naturally ventilated offices in Thailand was 0.45 clo which was higher than that in naturally ventilated residences.

3.1.2. Metabolic rate

The metabolic rate was calculated using the following hoc weighting factors: 70% at the moment of the survey and 30% proceeding the moment of the survey [37]. The frequency of distribution analysis for the metabolic rate revealed that the metabolic rate of the majority of the surveyed subjects was close to 1 met. The range of the metabolic rate was between 0.94 met to 1.3 met. Comparison between the three studies cannot be made due to the difference in the procedure of calculation used for the estimation, though the estimated values were quite close. It must be emphasized that all the procedures used to estimate metabolic rate in field studies provide rough information and the risk of error at its best is no less than $\pm 15\%$ [23].

3.1.3. Air temperature

Statistical analysis was repeated for the air temperature records to determine the mean and the range of the indoor air temperature and to examine the variation of air temperature over time during the conducted survey. Air temperature is one of the most recognized parameter in thermal comfort studies. A summary of the descriptive statistics is listed in Table 2.

The indoor mean air temperature of this study was 30.7 °C and the standard deviation was 1.47 °C. The air temperature values were almost normally distributed with the degree of skewness and kurtosis of 0.21 and 0.23, in that order. Approximately 95% of all air temperatures should lie between 27.81 °C and 33.68 °C. It can be observed that the temperature range in this study varies from 26.5 °C to 35.3 °C. Comparing the obtained values with others field studies in the hot-humid tropics revealed an important detail often omitted by researchers related to the maximum indoor air temperature records which will be discussed below.

The mean, minimum, maximum and standard deviation of the indoor air temperature recorded in the field thermal comfort study in Indonesia were respectively 29.5 °C, 26 °C, 32.6 °C, 1.45 °C [1]. The higher indoor air temperature recorded in the present study reflects the fact that the indoor air temperature measurements

were conducted in several residences constructed with different building materials subjected to various outdoor climatic conditions. The analyzed results were not surprising. The maximum indoor air temperature recorded by Feriadi and Wong [1] was 32.6 °C. Such conditions would be mostly typical of houses built with concrete or brick walls. The current data are consistent with this hypothesis and our previous long-term records of indoor temperatures in various houses built with different construction materials confirmed our hypothesis [24]. The authors only mentioned that the majority of the respondents surveyed were living in typical low rise houses designed to utilize natural ventilation but no further detailed information about the type of the building materials. The time factor was excluded as the highest percentage of their surveys were conducted in the afternoon. The difference in the mean indoor air temperature between the two studies was about 1.24 °C. Microclimate surrounded the location under investigation might also affect the indoor air temperature.

The estimated mean, minimum, maximum and standard deviation of the indoor air temperatures in the de Dear database for the conducted study in Singapore were 29.41 °C, 26 °C, 31.9 °C and 0.051 °C respectively. Here, even narrower range of indoor air temperatures was observed. In addition to the previous reasons stated above, the difference between both studies could be attributed to time factor. The higher percentages of votes recorded in their field study were between 6 p.m. and 9 p.m. Therefore, the difference in the mean indoor air temperature between both studies could be also attributed to outdoor climatic conditions reflected by the time factor. Table 3 shows sampling distribution of votes between the three studies at various time slots. Hourly records are also presented in Fig. 3. Feriadi and Wong [1] study was excluded because of the non-availability of more detailed information.

The recorded votes in the de Dear et al. [2] study were not uniform but rather mostly concentrated around specific time from 6 p.m. to 9 p.m., as opposed to the current study which recorded almost a uniform number of votes except before 9 a.m. or after 6 p.m. Admittedly, this study recorded the lowest number of votes from 7 p.m. to 9 p.m. and no record after 9 p.m. Also it should be noted that this study was carried out continuously every month for one whole year duration and each subject was visited only once. Therefore, the result of this study is more representative of thermal comfort field study during daytime.

Feriadi and Wong [1] carried out their investigation on April, May, June, November, December, and January. Their study was representative of the hot and rainy season but no information were reported about the number of records in each month and the hourly records for further examination. For de Dear et al. [2] field study, the survey was conducted during two months on August and September.

The large range of the indoor temperatures could be attributed as well to the variation of the indoor temperatures recorded during one whole year and thus covered larger indoor climatic conditions. Microclimates of the surrounding areas under investigation also could affect the indoor thermal environment as the outdoor

Table 2
Descriptive statistics of indoor air temperature.

Mean	30.74
Sample standard deviation	1.47
Minimum	26.50
Maximum	35.30
Confidence interval 95% lower	30.65
Confidence interval 95% upper	30.84

Table 3
Sampling distribution of votes at various time slots.

Time	Current study	Feriadi and Wong [1]	de Dear et al. [2]
	Malaysia	Indonesia	Singapore
06 a.m.–12 p.m.	279	97	42
12 p.m.–06 p.m.	521	292	237
06 p.m.–12 a.m.	87	137	304

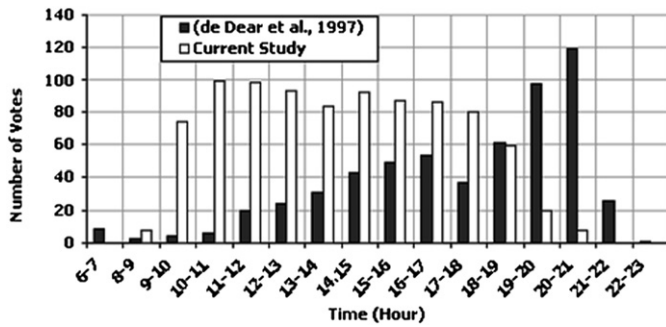


Fig. 3. Hourly sampling distribution of votes.

temperature may not necessary be the same as that reported by meteorological stations.

3.1.4. Globe temperature

The globe temperature provides useful detail about the effect of indoor long-wave radiation on indoor thermal comfort. The maximum difference between the globe temperature and air temperature in the present field study is 1.1 °C having a difference in terms of mean value of 0.21 °C. The range of the recorded globe temperature was 26.7–35.4 °C. The maximum difference between indoor air and globe temperatures in de Dear et al. [2] investigation was 0.9 °C. The difference between the mean values of these two parameters was 0.37 °C. Feriadi and Wong [1] did not summarize the results of the recorded globe temperature values but rather provided descriptive analysis for the mean radiant temperature. Overall, the three studies well agreed that air temperatures followed closely globe temperatures. The present results are in agreement with the conducted study by Sharma and Ali [9] in the development of the Tropical Summer index and by Webb [26] in the development of the named the Singapore index were the researchers found that the globe temperature agree closely with the air temperature and therefore globe temperature was subsequently ignored.

3.1.5. Relative humidity

Relative humidity in Kota Kinabalu is generally high within the mean monthly range of 79.5–82.2% [24]. This is the case of most maritime locations in the hot-humid tropics. Outdoor relative humidity generally decreases when air temperature increases [24]. This can be observed as well from climatic data of any location in Malaysia. The mean, maximum, minimum, and standard deviation of the measured relative humidity in the survey were 70.71%, 86.4%, 52.0%, and 6.4% respectively. These results are quite close with the reported values by Feriadi and Wong [1], whereas de Dear et al. [2] recorded slightly higher indoor relative humidity values. The standard deviations among the three studies are similar.

3.1.6. Air movement

Air movement is one of the most difficult environmental parameters to measure accurately and requires precise equipment as it is a highly variable parameter and can have more than one direction [13]. All the analyzed environmental parameters in this study were roughly normally distributed except for the indoor air movement records which were positively skewed. Therefore a transformation for removing the positive skew is squaring each of the air movement values [27]. This may provide insight of the reason behind squaring air movement in the elaboration of the thermal comfort indices in the previous studies such Singapore Index [26] and Tropical Summer Index [9]. The mean and the standard deviation of air movement after transformation were

0.2 m/s and 0.06 m/s, whereas the estimated mean value before normalization was 0.26 m/s. The maximum and minimum air movement records were 0 m/s and 1.5 m/s, respectively.

The mean air movement recorded by Feriadi and Wong [1] in their field study in Indonesia was 0.1 m/s. The minimum, maximum measurements and the standard deviation were 0 m/s and 1.2 m/s and 0.25 m/s both seasons included. The authors did not provide any information related to the test of normality. For the case of the de Dear et al. [2], the recorded mean, maximum, minimum and standard deviation were successively 0.2 m/s, 0.6 m/s, 0.1 m/s and 0.12 m/s. It can be observed that in the de Dear et al. [2] study; the maximum air movement record was very low compared to both studies. It must be emphasized that most of the subject under investigation were using fan. Overall, it can be stated that the recorded mean air velocity in the three studies was no more than 0.2 m/s. The mean air movement recorded in Thailand by Busch [17] was 0.33 m/s in naturally ventilated offices.

3.2. Preliminary analysis of ASHRAE thermal sensation votes

The frequency distribution of votes on ASHRAE thermal sensation scale from the number of observations is shown in Fig. 4. The vote distribution was found to be nearly normally distributed and the mean vote was 0.261 which is close to neutrality and the standard deviation was 1.39. The degree of skewness and kurtosis were –0.04 and 0.23, respectively. About 43.8% voted ‘neutral’, 20% voted between ‘cold’ to ‘slightly cool’ and 36.2% voted between ‘slightly warm’ to ‘hot’.

3.3. Expressing thermal sensation as a function of personal and environmental factors

The main objective of this section is expressing thermal sensation vote (TSV) as a function of personal and environmental factors to which the surveyed people were exposed. Correlation analysis between indoor air temperature (T_a , given in °C), relative humidity (RH, given in %), globe temperature (T_g , given in °C), water vapor pressure (P_a , given in pa) calculated from the combination of the air temperature and relative humidity according to ISO 7730 [7], air movement (v_a , given in m/s^2), metabolic rate (MET, given in met), clothing insulation (I_{cl} , given in clo) were selected for the preliminary analysis as independent variables and thermal sensation votes as dependent variable. Results are summarized in Table 4.

Among all the predicted variables, the correlation between air temperature and thermal sensation vote tended to be higher than the remaining variables except for the globe temperature which has almost similar correlation coefficient with the thermal sensation

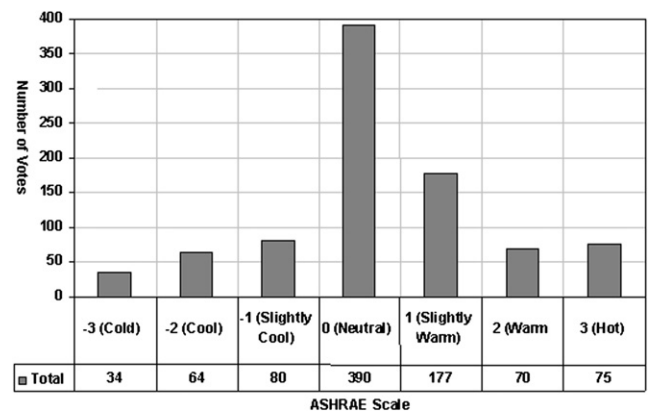


Fig. 4. Relative frequency of ASHRAE thermal sensation votes.

Table 4
Correlation coefficients between selected variables.

Variables	TSV	I_{cl}	MET	T_a	T_g	RH	P_a
TSV	1.000						
I_{cl}	0.014	1.000					
MET	0.013	0.078 ^a	1.000				
T_a	0.417 ^b	−0.020	−0.017	1.000			
T_g	0.415 ^b	−0.014	−0.013	0.994 ^b	1.000		
RH	−0.299 ^b	0.062	0.016	−0.696 ^b	−0.695 ^b	1.000	
P_a	0.119 ^b	0.059	0.006	0.311 ^b	0.305 ^b	0.464 ^b	1.000
Sqrt(v_a)	−0.108 ^b	0.025	−0.026	0.100	0.088 ^a	−0.099 ^b	−0.007

^a Critical value 0.05 (two-tail).

^b Critical value 0.01 (two-tail).

vote. This is a warning sign that collinearity exists between air temperature and globe temperature. This means that one of the two variables is redundant [27]. Thus, it is only necessary to include one of the two variables in the model for comfort temperature prediction. This might partly occurred because occupants have more freedom to select pleasant spaces and the least subjected to the heat induced by long-wave radiations compared to offices. The presence of furnishings can also greatly increase the surface area for heat exchange with the room air, so reducing the air-radiant temperature difference as reported by Humphreys et al. [22]. Thus, long-wave radiation seems to be a negligible component in this study. Additionally, linear regression requires that the independent variable should be free from error measurements [27]. Globe temperature is not only subjected to error measurements due to the equipment as the case of air temperature but it is further subjected to additional measurements error due to the use of ping-pong or 150 mm diameter black painted globe thermometer. Therefore, the parameter air temperature was kept and the parameter globe temperature was discarded.

In Malaysia as reported in Section 3.1.5, relative humidity generally decreases when air temperature increases. This means when air temperature is high, relative humidity is low but occupants' thermal sensation increases because air temperature has more effect on occupant thermal sensation than the effect of relative humidity. Nicol [36] reported that relative humidity of the air has been used in most studies of thermal comfort which is highly dependent on air temperature especially at high temperatures. The author stated that the water vapor pressure should be a good predictor of the evaporative heat loss because the driving force for heat loss by evaporation is the difference between the saturated water vapor pressure at the skin temperature and in the surrounding air according to the author. In this investigation, relative humidity correlates better with TSV ($r = -0.299$) than water vapor pressure ($r = 0.119$) as shown in Table 4 and it also has higher correlation coefficient with air temperature ($r = -0.696$) compared to water vapor pressure ($r = 0.311$) in the existing correlation though both were significant at 0.01 level. Therefore, relative humidity as an independent variable was discarded. It is necessary to highlight that despite it was not possible to distinguish between the separate effect of air temperature and relative humidity level on occupants' thermal comfort, humidity may have a strong effect on thermal sensation at warm to hot temperatures. The results from this investigation are only applicable for the range of the indoor environmental and personnel parameters under consideration. Extrapolation and interpretation of the result beyond that range for further generalization is not valid specifically when using regression models.

The squared air movement has a weak correlation with TSV but significant and the correlation between the same parameter with globe and relative humidity are significant. This means that this parameter cannot be used by itself to predict the TSV but might work better when combined with some of the selected parameters.

The weak correlation between the squared air movements and occupants' thermal sensation may also be attributed to the very narrow range of the recorded mean air movement at various indoor air temperatures [38]. Thus the rate of change between air movement and occupant thermal sensation might not be considerable.

Metabolic rate and clothing insulation have lower correlation coefficients than air movement and not significant. The weak and insignificant correlation between clothing insulation and occupants' thermal sensation can also be attributed to the negligible rate of change of clothing insulation at various indoor air temperatures. This most probably happened because clothing adjustments took place at the low range of indoor air temperature records thus further decrease in clothing insulation values with the increase of indoor air temperatures may not be possible because it reached a critical lowest value. For the metabolic rate, the recorded values were almost close to one met. The metabolic rate recorded in this study was almost a constant rather than a variable to be included in a regression model. In the dear database, it was also found that mean metabolic rates in the overall worldwide field studies were tightly grouped in the 1.1–1.4 met unit range regardless of the outdoor temperature. This may occurred because of the preset requirements set in most of thermal comfort field studies by allocating at least 20–25 min to respondents prior starting records of occupants thermal sensation.

Among the remaining variables, there were 16 possible combinations with the air temperature included in each regression. Table 5 shows all the possible combinations between selected variables and thermal sensation vote with their P -values. The adjusted coefficient of determination is the modified coefficient of determination (r^2) to account for the number of variables and the sample size. In order to determine the best five models among the 16 possible combinations, stepwise regression was considered. From the obtained results, it appears that adding more variables does not improve much the regression model, as air temperature does relatively fairly well by itself. Adding air velocity values improve marginally the correlation. However, because of the uncertainty in the air velocity measurements it was discarded as the effect of the measurements error become more unpredictable. Metabolic rate should be excluded because it is almost a constant value.

Newsham and Tille [28] cited over 15 studies were the performance of air temperature, operative temperature, or some other composite temperatures were very similar in thermal sensation predictions on the ASHRAE scale and on a Bedford Scale. This was also observed in the field study conducted by Karyono [18] in

Table 5
Possible combinations of the selected variables.

N_{var}	I_{cl}	MET	T_a	P_a	Sqrt(v_a)	Adj. r^2	r^2
1 ^a			7.44E−37			0.173	0.174
2 ^a			9.19E−40		1.37E−06	0.195	0.197
2	0.4716		6.89E−37			0.173	0.175
2		0.5336	7.21E−37			0.173	0.175
2			3.23E−34	0.7175		0.172	0.174
3 ^a	0.3879		7.94E−40		1.23E−06	0.195	0.198
3			3.60E−37	0.5733	1.26E−06	0.194	0.197
3		0.6080	9.31E−40		1.48E−06	0.194	0.197
3	0.5008	0.5701	6.82E−37			0.172	0.175
3	0.4556		2.73E−34	0.6804		0.172	0.175
3		0.5309	3.08E−34	0.7119		0.172	0.175
4 ^a	0.3658		2.75E−37	0.5322	1.11E−06	0.194	0.198
4	0.4093	0.6553	8.23E−40		1.33E−06	0.194	0.198
4		0.6037	3.60E−37	0.5695	1.36E−06	0.194	0.197
4	0.4843	0.5683	2.68E−34	0.6777		0.171	0.175
5 ^a	0.3864	0.6528	2.82E−37	0.5307	1.21E−06	0.193	0.198

N_{var} : Number of variables.

^a is referred to all best combinations.

Jakarta when using the operative temperature and air temperature in their prediction of indoor neutral temperature.

When evaluating worldwide thermal comfort database known by de Dear database and the SCATs database, Humphreys et al. [22] also found that the use of air temperature or operative temperature as independent variables are relatively superior to the following indices New effective temperature (ET^*), standard effective temperature (SET) and (PMV). The use of air temperature as the single parameter for the prediction of neutral temperature is simpler, safer, and relatively accurate and it can be easily understood. Adding to that, this will reduce the error that may be induced from the measurements of wind velocity and globe temperature which compensate the marginally improvement when combining the two parameters or when using other indices.

3.4. Prediction of the indoor neutral temperature

In this investigation, linear regression was selected because it is widely used method for neutral temperature prediction [28], though, probit analysis is well known method for the prediction and it seems to provide similar outcome, and studies that have used both methods have found very similar results [28]. Further investigation in probit and logit regression for neutral temperature prediction will be the objective of another publication in the near future. For the prediction of the indoor neutral temperature, the relationship between thermal sensation votes (TSVs) and the indoor air temperatures (T_a) was first examined by applying least squares linear regression. The regression line was drawn through the data and it was highly significant ($F = 186.65$, P value = $1.04.10^{-38}$; $r^2 = 0.174$; $n = 890$).

$$TSV = 0.395T_a - 11.875 \quad (1)$$

The regression mean squares value obtained from the least squares regression was 297.61, whereas the residual mean squares value was 1.59. This indicates the existence of linear relationship between indoor air temperatures and thermal sensation votes. The coefficient of determination of the regression equation of the present study was about 17.4%. Statistically this means that temperature explains 17.4% of the variation of occupants' thermal sensation. The coefficient of determination of the developed model was small though quite widely observed in thermal comfort field studies specifically when using all dataset. The coefficients of determination reported by some investigators in thermal comfort studies were 0.08, 0.20, 0.14, 0.18, and 0.32 [1,20,29–31]. Newsham and Tiller [28] suggested that this occurs due to the narrow range of temperatures to which the occupant is exposed compared to what the human body can physiologically adjust. Some researchers attributed the results to other omitted psychological parameters, personality and physical environment parameters such lamp type, ceiling height, occupancy density [28]. The obtained result may also be ascribed to the way of using linear regression with measurements of thermal sensation of several individuals for repeated indoor air temperatures with people having different thermal sensation. Thus the coefficient of determination will be weakened due to the variation in the votes of occupants subjected to similar indoor temperature rather than to the correlation between air temperature and thermal sensation.

Several investigators preferred using mean thermal sensation values calculated for each temperature bin instead of using the whole dataset for the determination of the indoor neutral temperature from the least squares linear regression. Though, some experts in statistics do not recommend the use of averages values as this not only will suppress individual variation but it may inflate the correlation coefficient [32,33]. Busch [17] reported that using

binned mean responses or all responses, the slope and intercept remain the same. Fig. 5 shows the relationship between mean thermal sensation within each successive 0.5 temperatures bin and mean indoor air temperature within each 0.5 bin. Also shown in the figure are the 95% confidence interval and 95% prediction interval for the regression line.

The regression line was drawn through the data and it was significant ($F = 238.13$, P value = $1.97.10^{-11}$; $r^2 = 0.933$; $n = 19$).

$$TSV_{(mean)} = 0.383T_{a(mean)} - 11.572 \quad (2)$$

The coefficient of determination of the regression is close to one although the slope and the intercept are very similar to that obtained from the previous approach. Many experts in thermal comfort field studies [10,20] applied the above approach in predicting indoor neutral temperature but with some alterations. In these studies fits of the regressed model were weighted by the number of votes according to the number of observations associated with each mean [10,17,20,28]. de Dear and Fountain [20] pointed out that this would minimize the effect of relatively large residuals at the tails of the sample where there is less frequently encountered temperature at most distant from the mean. This is known in statistics as weighted least squares with replicates or near replicates [34]. Replicate observations are made at each mean indoor air temperature bin. This will be referred to it as the third approach. The fourth approach for the prediction of the indoor neutral temperature is similar to the second approach but the mean values were estimated from every 25 votes in ascending temperature order. This method was used in the development of the Humphreys windows opened algorithm [35]. This approach may provide more representative mean indoor temperature values and therefore the regression line was expected to be more accurate.

Table 6 summaries the outcomes of the four approaches. Table 6 shows that all the applied approaches provides broadly identical neutral temperatures of $30.1\text{--}30.2\text{ }^\circ\text{C}$ having a confidence interval of ± 0.1 to $\pm 0.2\text{ }^\circ\text{C}$. The slope of the regressed models as depicted in Table 6 indicates that an increase of air temperature of about $2.5\text{ }^\circ\text{C}$ is equivalent to an increase in one unit of thermal sensation. This implies that the comfort range varied within one unit change in the thermal state from neutrality to either slightly warm or slightly cool. Surprisingly, this is equivalent to the predicted confidence interval from Eq. (1) for an individual.

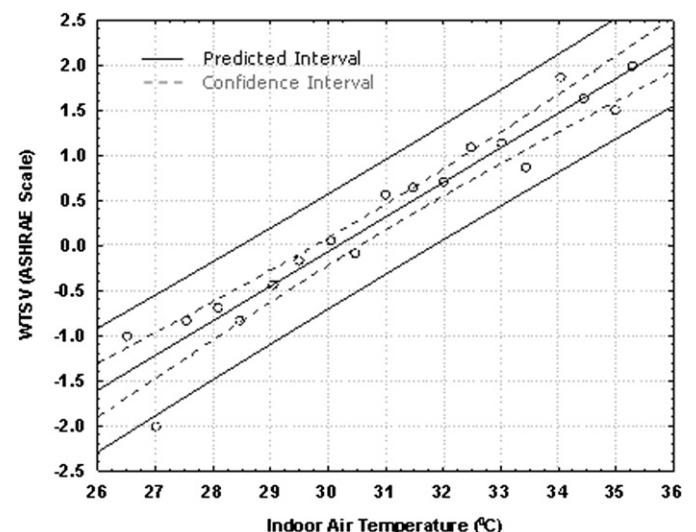


Fig. 5. Occupants votes vs. indoor temperature.

Table 6
Neutral temperature using various approaches.

Approach	Regression equation	Neutral temperature	r^2 (P value)
Approach 1	$TSV1 = 0.3957T_a - 11.875$	30.1 CI: (−0.1)–(0.1) PI: (−2.5)–(2.5)	0.17 (≤0.000)
Approach 2	$TSV2 = 0.3837T_{a(\text{mean})} - 11.572$	30.2 CI: (−0.1)–(0.2) PI: (−0.6)–(0.6)	0.93 (≤0.000)
Approach 3	$TSV3 = 0.3937T_{a(\text{mean w})} - 11.810$	30.1 CI: (−0.1)–(0.1) PI: (−0.3)–(0.3)	0.93 (≤0.000)
Approach 4	$TSV4 = 0.4137T_{a(\text{mean } 25)} - 12.433$	30.1 CI: (−0.1)–(0.1) PI: (−0.6)–(0.6)	0.82 (≤0.000)

The predicted interval when using all responses (± 2.5 °C) was wider than the case of other approaches having a maximum wide of ± 0.6 °C. This statistically indicates that the occupant thermal sensation of an individual under neutrality lies between 27.6 °C and 32.6 °C and the thermal sensation vote of the population is within 30 °C and 30.2 °C based on the first approach. From the analysis, it is apparent that the predicted interval when using the first approach is more appropriate. This is because averaged values repress individuals' variation by reducing the predicted interval as reported above, though the coefficient of determination might not accurately represent the strength of the correlation. However, decisive conclusions cannot be made before checking the variance and the normal plots of residuals.

The obtained results from the applied approaches are plotted in Fig. 6 for the temperature range of 26.5–35.5 °C. The present study confirmed that least squares linear regression gave similar outcomes regardless of the procedure for the prediction of the indoor comfort temperature.

The first, the third, and the fourth approaches listed in Table 6 were selected for analyzing the assumptions underlying linear regression analysis. The tested assumptions are to check whether or not residuals are normally distributed and the assumption of linearity is satisfied and most importantly to examine if the assumption of constant variance is valid. If the former assumption is violated, confidence interval and predicted interval may not be accurate. Approach 3 was discarded due to the similarity of the results and procedure with approach 2 and also because it is easier to check all the requirements using mean values rather than weighted mean values.

When the plotted residuals vs. predicted values and the normal score of the evaluated approaches were analyzed, it was found that that the distributions of the residuals were roughly normal.

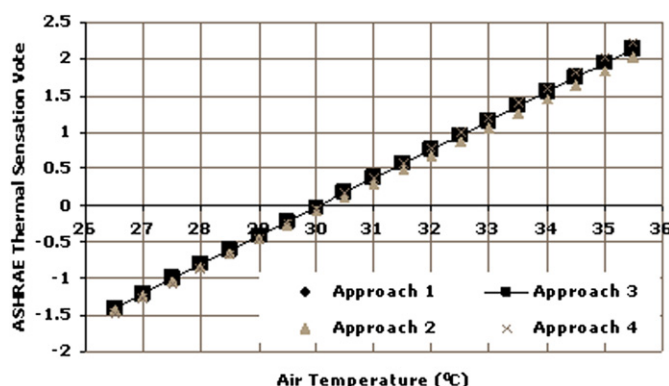


Fig. 6. Comparison between different approaches.

However, the plots of fitted values against residuals suggested that the assumption of linearity was satisfied, whereas the assumption of constant variance was mostly violated for approach 1 and 4. The increasing variability for these two approaches was apparent at the middle of the predicted votes around the neutrality. Therefore, the dataset has violated the assumption of constant variance. This was attributed to the large number of occupants who voted differently around the mid-range of indoor air temperatures due to the variability of their thermal sensation compared to those voted at the extreme air temperatures having mostly similar thermal sensation. Thus, when the averaged values were used this effect disappeared. The calculations of significance and the estimated confidence intervals and the prediction intervals when applying the first approach and the fourth approach might not be accurate, and might be wrong to some extent. However, the violation of this assumption will not alter the parameter estimates [27]. Therefore, from the whole analysis, it can only be concluded that the predicted mean neutral indoor temperature was 30.2 °C $\pm$ 0.2 °C for the population under investigation but the predicted interval for one individual is not known because of the violation of the assumption of constant variance. Also it was found that the increase of the indoor air temperature by about 2.5 °C was equivalent to an increase in one unit of thermal sensation. All the methods provided almost similar neutral temperature regardless of the violation of some of the least squares linear regression assumptions.

When evaluating worldwide thermal comfort studies from de Dear and the SCATs databases, Humphreys et al. [22] observed that the average mean regression gradient in the conducted surveys was approximately 0.4 considered as standard slope regression which was recommended to be used rather than the empirical value for each individual survey. Further, the authors observed that is possible to predict the neutral temperature for thermal neutrality of a group from Eq. (3).

$$T_{n(\text{mean})} = T_{a(\text{mean})} - TSV_{(\text{mean})}/b \quad (3)$$

where (T_n) is the mean indoor neutral temperature, (T_a) is the mean indoor air temperature recorded during field survey, whereas (b) is the standard regression gradient equal to 0.4.

The mean indoor air temperature of all responses in the present study was nearly 30.7 °C and the mean thermal sensation vote $TSV_{(\text{mean})}$ was almost equal 0.3. From Eq. (3), the calculated mean indoor neutral temperature is 30 °C which is in close agreement with the present finding.

4. Discussions

4.1. Comparison to previous field studies

The present investigation provides the possibility for comparison between results among studies conducted in naturally ventilated residences specifically in the hot-humid tropics. The indoor neutral temperature found in the present investigation was higher by about 1.2 °C compared to the indoor neutral temperature determined by Feriadi and Wong [1] in their field study (28.8 °C air temperature or 29.2 °C operative temperature) for naturally ventilated houses in Jakarta.

The indoor neutral temperature predicted by de Dear et al. [2] in naturally ventilated residences in Singapore was about 28.5 °C (having operative temperature as an independent variable) which is lower by nearly 1.5 °C compared to the present investigation. It appears that the indoor neutral temperature is very close to the mean indoor air temperature under investigation and the indoor range affects the predicted indoor comfort temperature. In addition to the wide range of indoor temperature recorded in the present

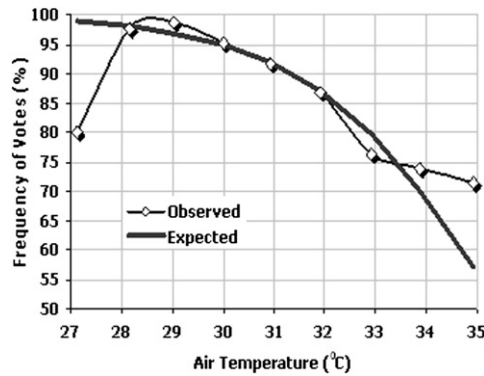


Fig. 7. Thermal acceptability (logit regression).

investigation compared to Jakarta and Singapore studies which seem to have effect on neutral temperature prediction, the discrepancy might as well be attributed to the slight increase of the mean air movement recorded in this study compared to Jakarta investigation. It must be emphasized that in the present investigation at least 80% of occupants were using fan most of the time [38]. The heavy reliance on the use of fans to relieve thermal discomfort has most probably enabled the neutral temperature to be higher than previous studies due to the convective cooling from fan. In addition to that having control over the fans may also affect occupants' thermal sensation toward indoor temperatures. In a field study conducted at Berkeley civic centre in office buildings, Brager et al. [10] found that in the warm season, people with a higher degree of control over the windows were comfortable at warmer temperatures by about 1.4 °C than the group with lower levels of control. This highlights the importance of having more control is among the reasons of selecting higher neutral temperature. One more possible explanation is that the annual outdoor temperature in Singapore is lower than the case of Malaysia by 1 °C so it would be expected that the indoor neutral temperature in Malaysia will be higher by about 1 °C than in Singapore [25,39]. Though, microclimate of the surveyed locations may also affect the indoor neutral temperature predictions. The discrepancy between results might also be ascribed to differences in the outdoor air temperatures during the period under investigation and to the differences in habits and climatic parameters.

Overall compared to office buildings, the predicted indoor neutral temperature in naturally ventilated buildings was higher than that predicted in offices and classrooms in the hot-humid tropics. Feriadi and Wong [1] attributed the difference to the availability of more controls in residences which allow the occupants to make their indoor environment more comfortable such changing clothes, getting more drinks, switching the fan on and

others. It must be emphasized that temperatures higher than 30 °C was found to be comfortable if occupants were using fan [36]. The detailed data on occupants' behavior in their dwelling in Kota Kinabalu should be subjected to thorough analysis and discussion in a future paper.

4.2. Thermal acceptability scale

Thermal acceptability range in the present analysis will be determined based on the percentage of the respondents voting 'Acceptable' on thermal acceptability scale. It is stipulated in ASHRAE 55 [6] as the condition in which at least 80% of occupants will find the environments thermally acceptable. Despite thermal acceptability scale has a binary answer 'Acceptable' (1); 'not acceptable' (0), and therefore appears to be the most suited scale for probit or logit analysis compared to ASHRAE and McIntyre scales, indoor thermal acceptability range cannot be predicted by using logit or probit analysis. This can be illustrated by the plot shown in Fig. 7 in which it can be observed that the curve made from the predicted votes based on thermal acceptability scale from logit regression is not increasing with the raise of the indoor air temperature but rather has an optimum value within the range under consideration. For such condition probit and logit is not suitable. When looking at many publications beyond thermal comfort studies, this observation is further emphasized and many others [40–42]. The percentage of distribution of votes on thermal acceptability scale is shown in Fig. 8. This reveals that over 80% of occupants found the indoor air temperature acceptable within the indoor temperature range of about 27–32.5 °C.

5. Conclusions

This paper of thermal comfort explored the indoor comfort temperature prediction and evaluation of residences in the Kota Kinabalu area from the ASHRAE seven-point scale. The mean thermal sensation of the subjects under investigation showed a heightened sensitivity to indoor temperature represented by a gradient of one sensation unit per 2.5 °C change in indoor air temperature. The neutral temperature was determined using various approaches. The predicted neutral temperature using least squares linear regression was found to be nearly 30 °C regardless of the adopted approach. From statistical analysis, it was concluded that the predicted indoor neutral temperature in residential buildings was 30.2 °C ± 0.2 °C for the population under investigation but the predicted interval for one individual is not known because of the violation of the assumption of homogeneity of variance when using all dataset.

Based on the present investigation, it seems to be good to have the widest possible range of the indoor temperature of the location under investigation for thermal comfort prediction. In addition both the dependent and the independent variables should be normally distributed. Further analysis about thermal comfort in wooden houses as opposed to concrete houses is highly recommended as it may not be similar.

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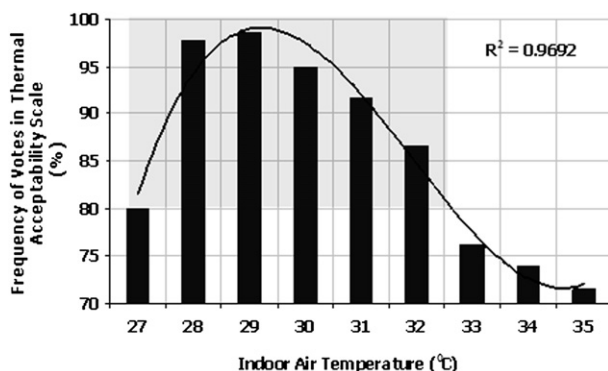


Fig. 8. Thermal acceptability (frequency distribution).

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